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Design and Simulation of a PID Controller for Precise DC Motor Speed Control

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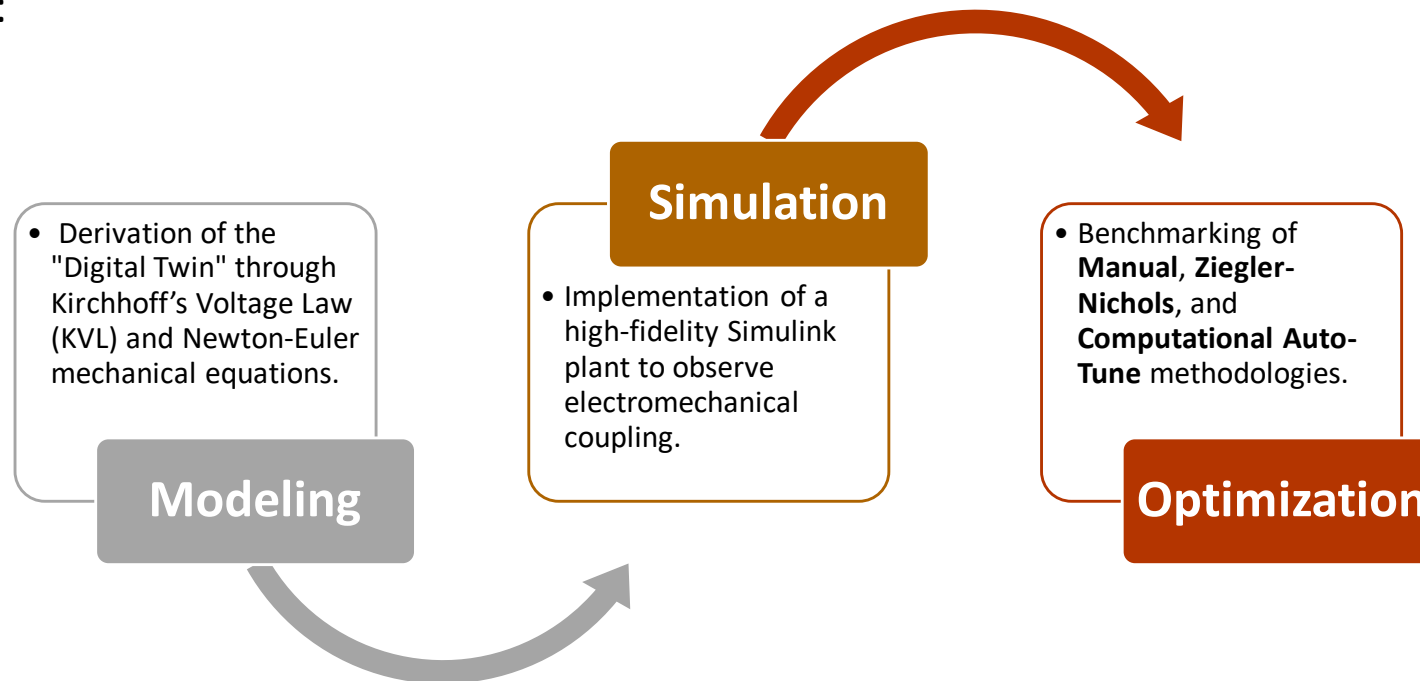


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The Problem Statement: Traditional open-loop DC motor systems are inherently susceptible to steady-state error and load disturbances. To achieve industrial-grade precision, a deterministic model must be synthesized with a robust control law.

Scientific Methodology:



Key Findings: Computational optimization yielded a stable response with **<10% overshoot** and **zero steady-state error**, even under dynamic load disturbances.





❖ Subsystem Decomposition

- Separation of the plant into the **Armature Circuit (Electrical)** and the **Rotor Dynamics (Mechanical)**.

The Electrical Domain:

- Implementation of the first-order differential equation:

$$V(t) = L \frac{di}{dt} + R \cdot i(t) + e_b(t).$$

- Integration of current (i_a) to determine the electromagnetic torque (T_m).

The Mechanical Domain:

- Implementation of the torque balance:

$$J \frac{d\omega}{dt} + b \cdot \omega(t) = T_m(t) - T_L(t).$$

- Modeling of the inertial resistance (J) and viscous damping (B).

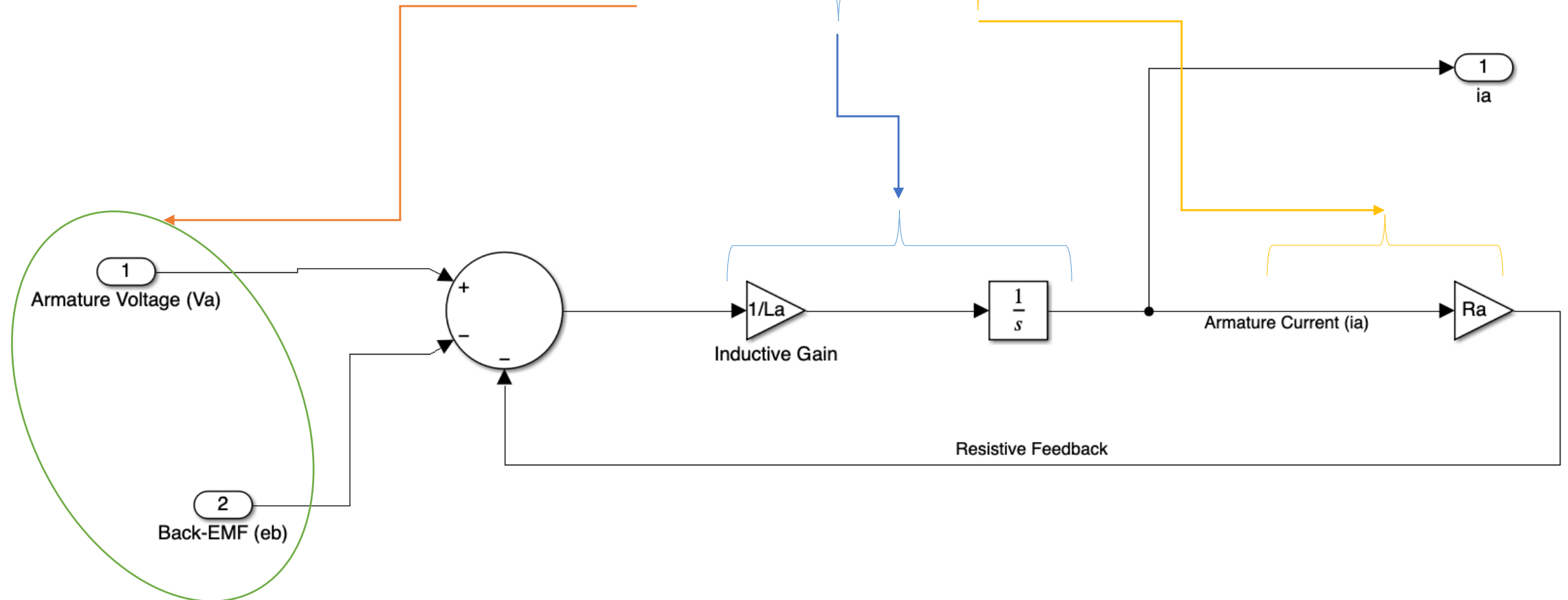
Inter-domain Coupling:

- **Torque Constant (K_t)**: Converting current into mechanical energy.
- **Back-EMF Constant (K_b)**: The feedback of velocity back into the electrical potential.



Back-EMF: $e_b = K_b \cdot \omega$ (Back-EMF is proportional to speed).

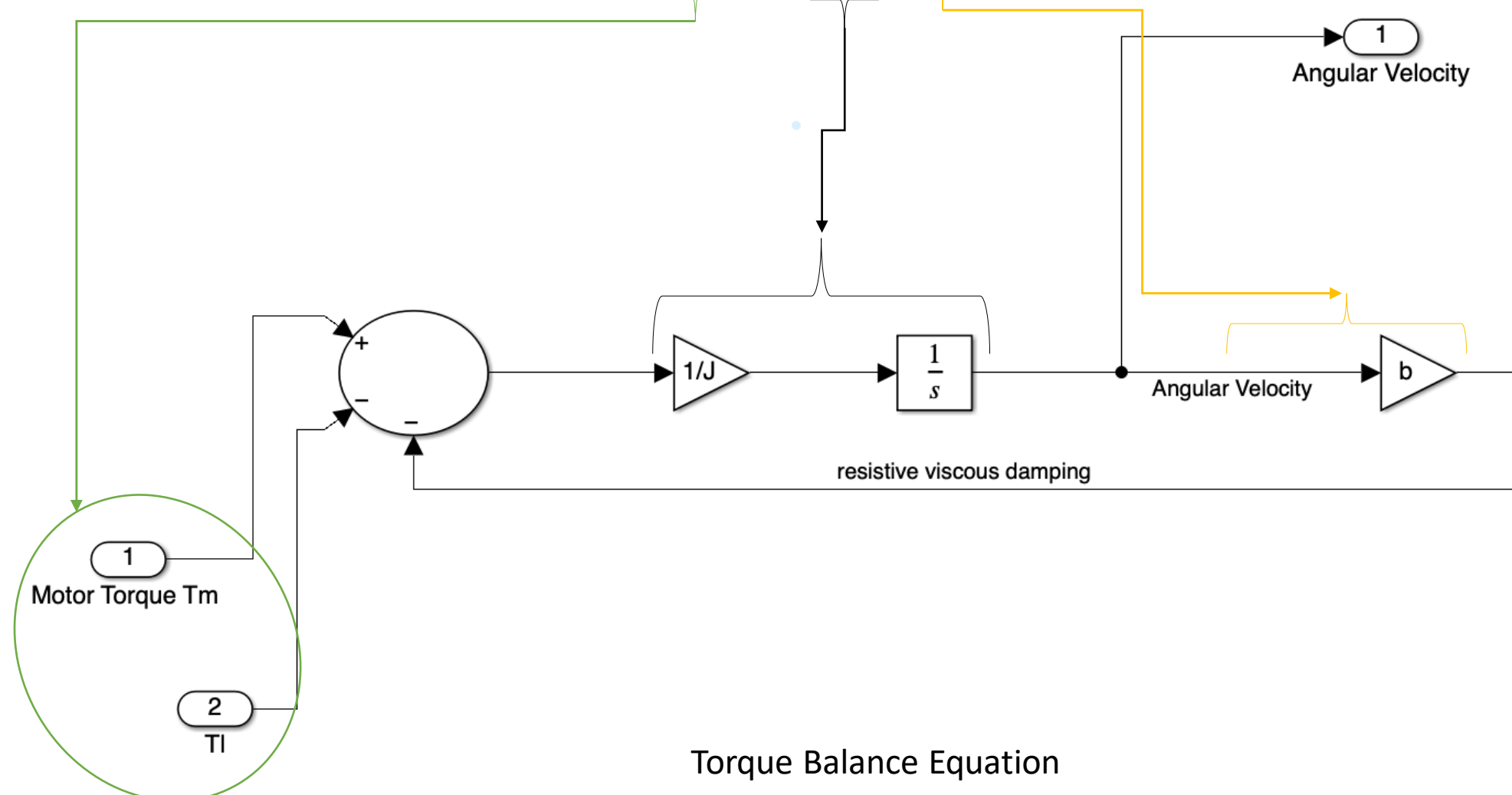
$$V_a - e_b = L_a \frac{di_a}{dt} + R_a i_a$$



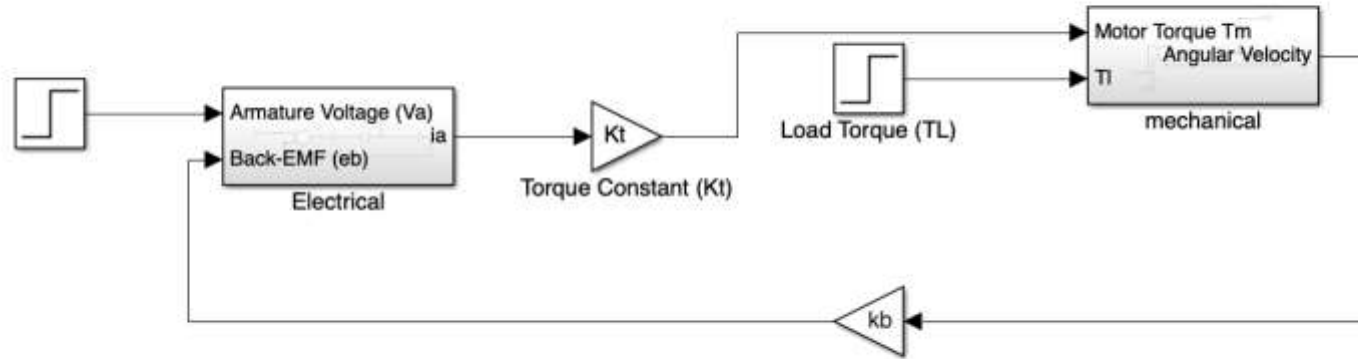
Kirchhoff's Voltage Law

Electromagnetic Torque: $T_m = K_t \cdot i_a$
(Torque is proportional to current).

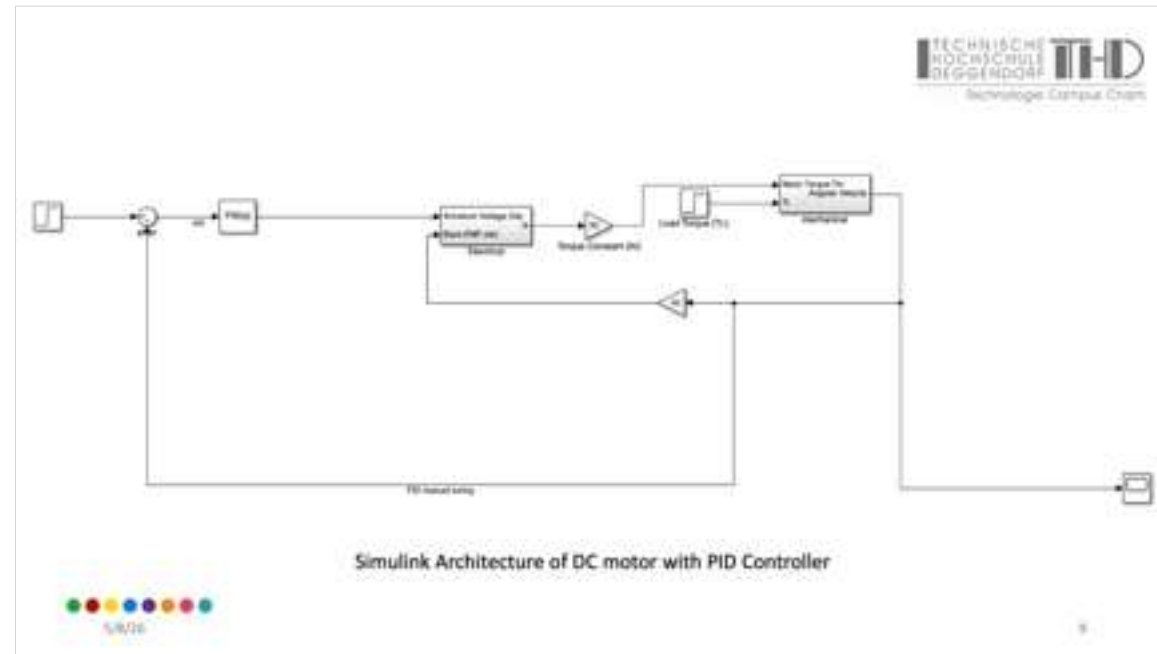
$$T_m - T_L = J \frac{d\omega}{dt} + B\omega$$

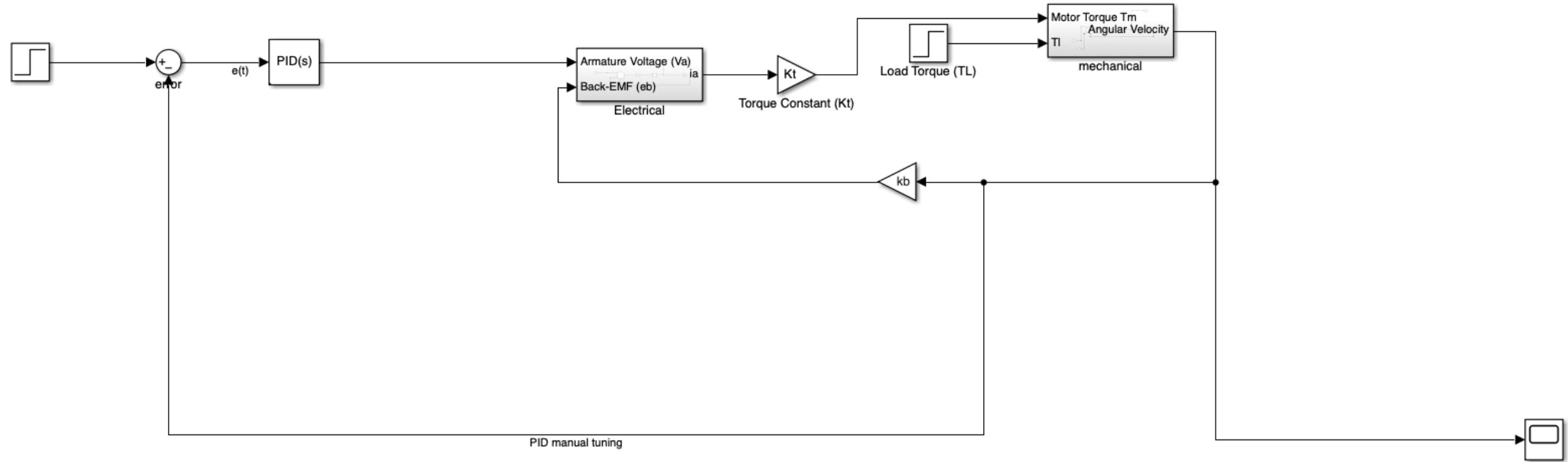


Torque Balance Equation



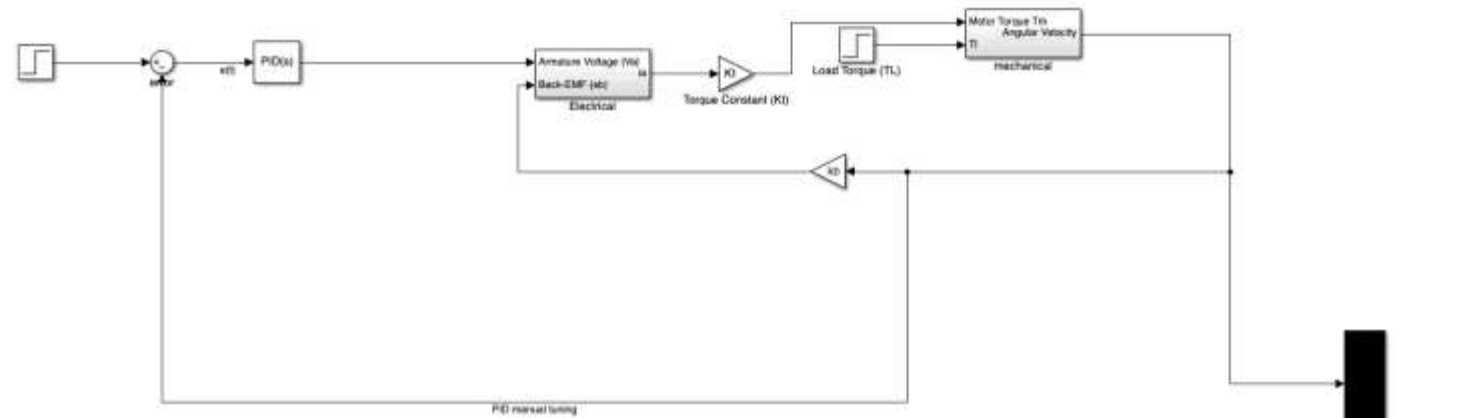
Detailed Simulink Architecture of DC motor



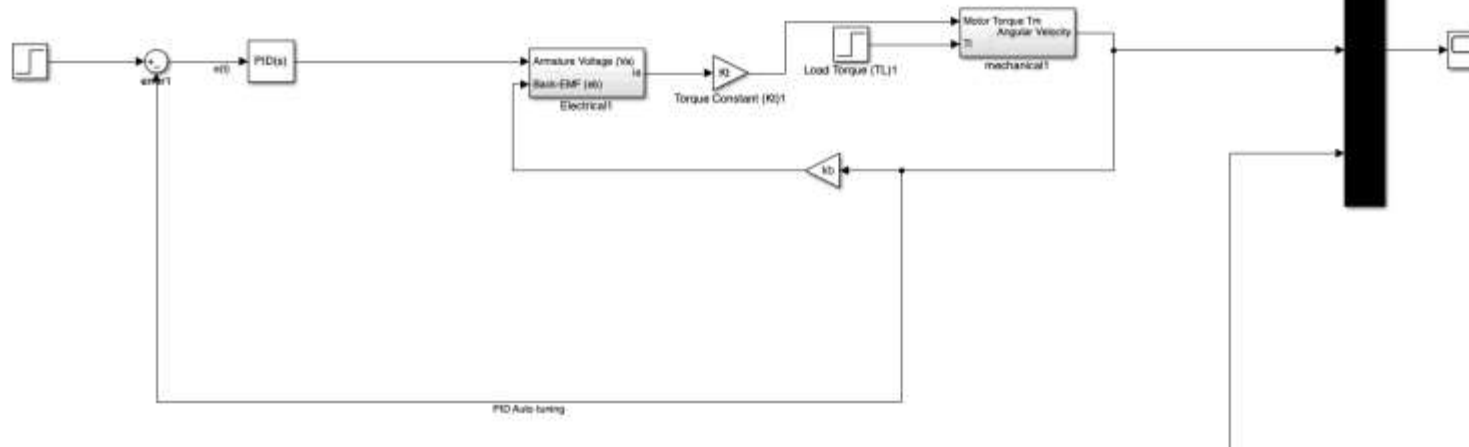


Simulink Architecture of DC motor with PID Controller

PID Manual Tuning



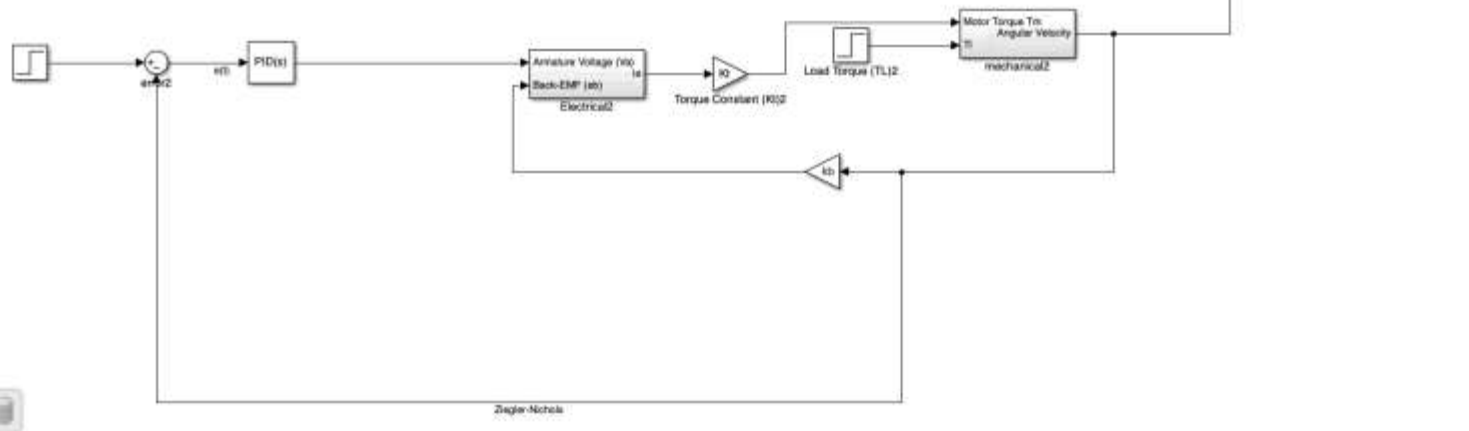
PID Auto-Tuning



Ziegler-Nichols



6/22/2026





Baseline Characterization (Open-Loop Response)

Analysis of the plant response to a unit step input (V_a) without a feedback loop or PID intervention.

Transient Behavior:

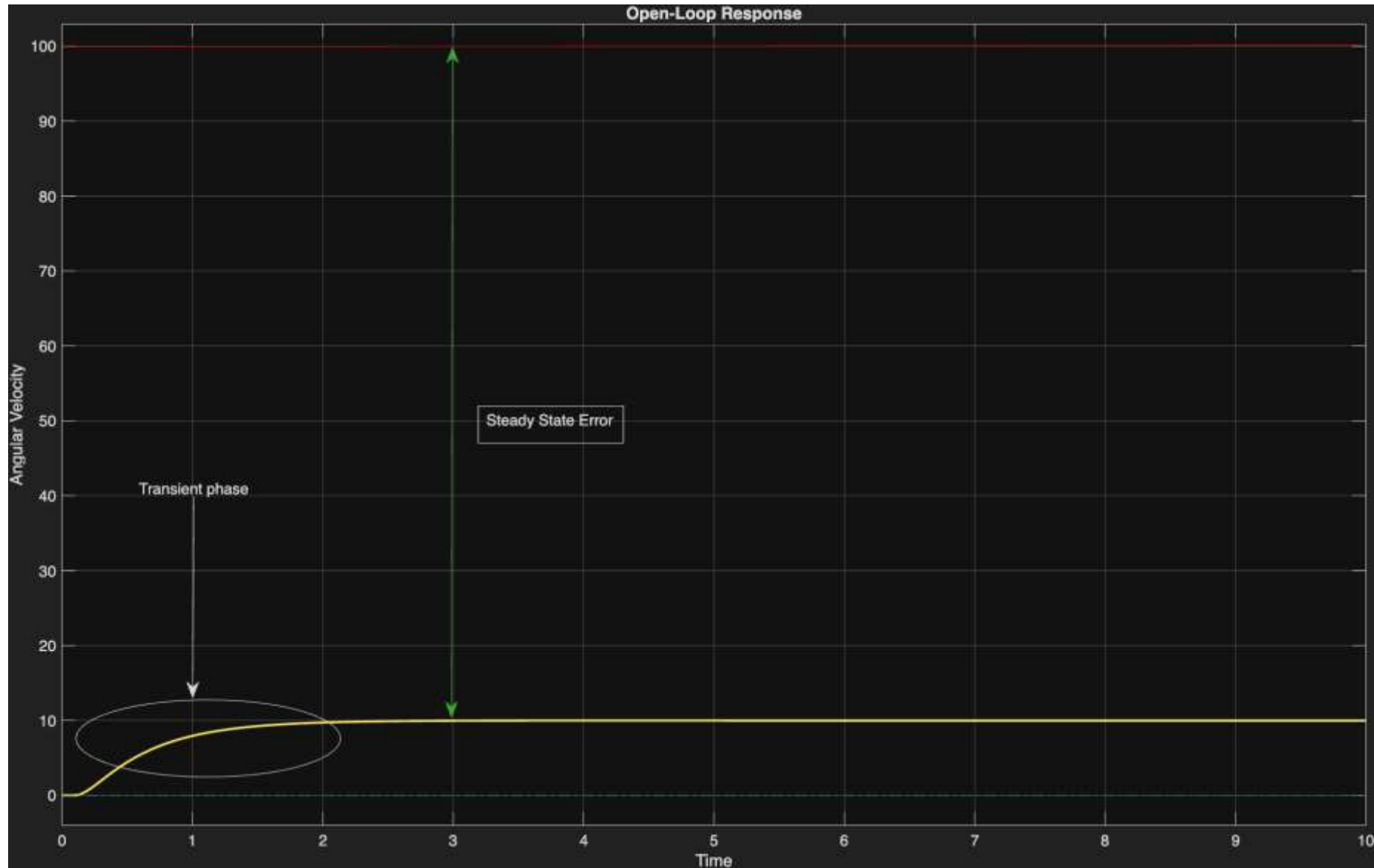
- The system exhibits a classic first-order or overdamped second-order exponential rise.
- The slope is governed by the electromechanical time constant $\tau_m \approx \frac{RJ}{K_t K_b}$, representing the natural "sluggishness" of the rotor inertia.

The Steady-State Offset (Problem Statement):

- The output velocity settles at a value determined purely by the DC Gain of the plant.
- **The Critical Gap:** There is a significant **Steady-State Error (e_{ss})**. The motor "drifts" to an arbitrary speed rather than following a precise command.

Sensitivity to Externalities

- In this state, any change in friction (B) or a sudden load torque (T_L) causes an unrecoverable drop in velocity, as there is no "correction signal" to increase voltage.





The PID Control

The system transition from Open-Loop to Closed-Loop is governed by the continuous-time control law:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Proportional (K_p) — The Present:

Generates a corrective signal proportional to the current error magnitude ($e(t)$). It provides the primary "drive" but is mathematically incapable of eliminating steady-state offset alone.

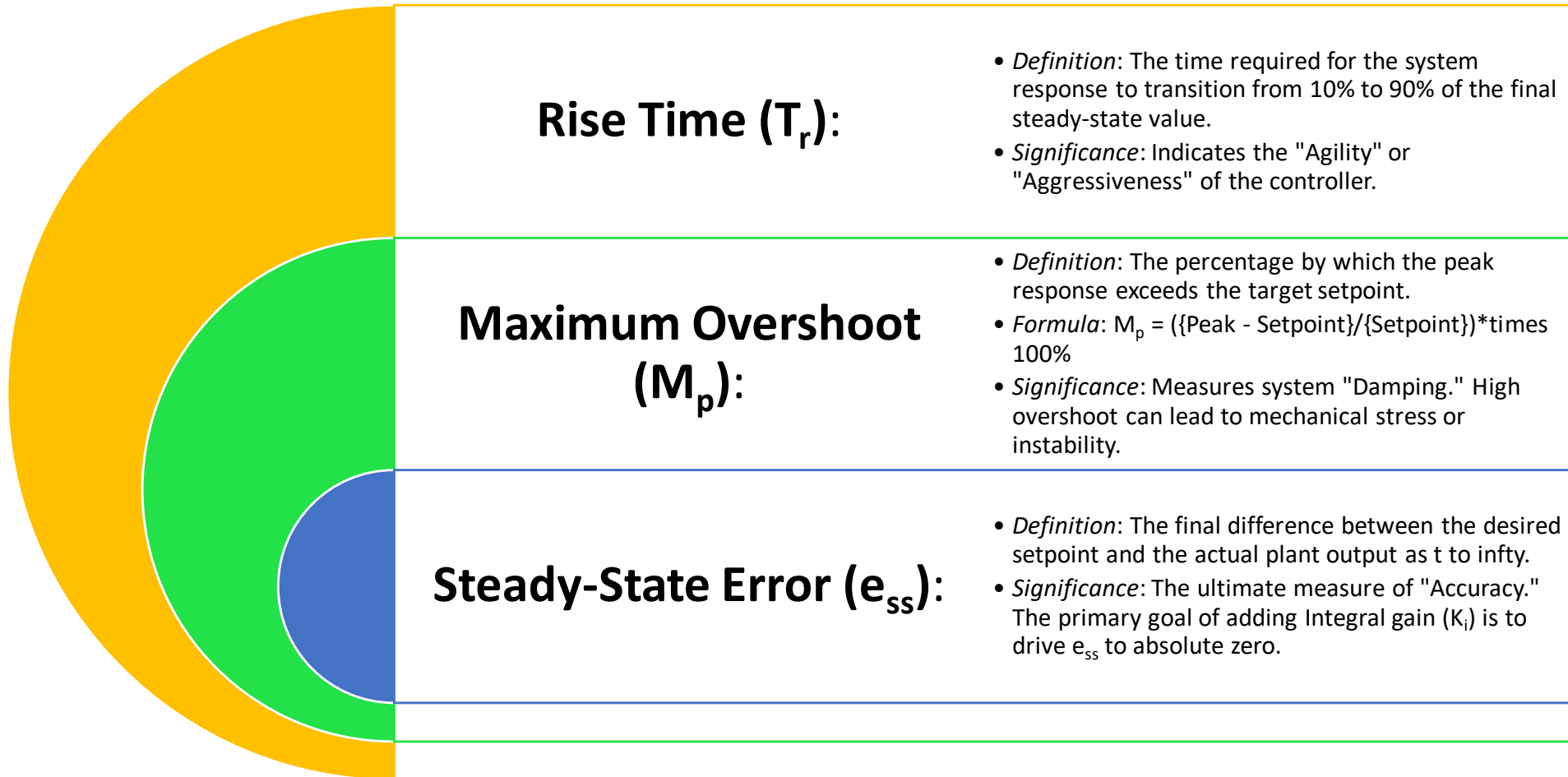
Integral (K_i) — The Past: Integrates the error over time to eliminate residual offsets. This is the "memory" of the controller that forces the motor to reach exactly 100 rad/s.

Derivative (K_d) — The Future: Predicts future error trends by calculating the rate of change. It acts as a mechanical "damper," preventing the aggressive Proportional term from causing excessive overshoot.

The Error Input ($e(t)$): Defined as the instantaneous difference between the Setpoint ($r(t)$) and the Process Variable ($y(t)$): $e(t) = r(t) - y(t)$.



To evaluate the efficacy of the PID control law, three primary transient and steady-state metrics are utilized:



The Proportional-Only (P) Experiment: Initial tuning began by setting $K_i = 0$ and $K_d = 0$, focusing solely on the impact of the Proportional gain (K_p).

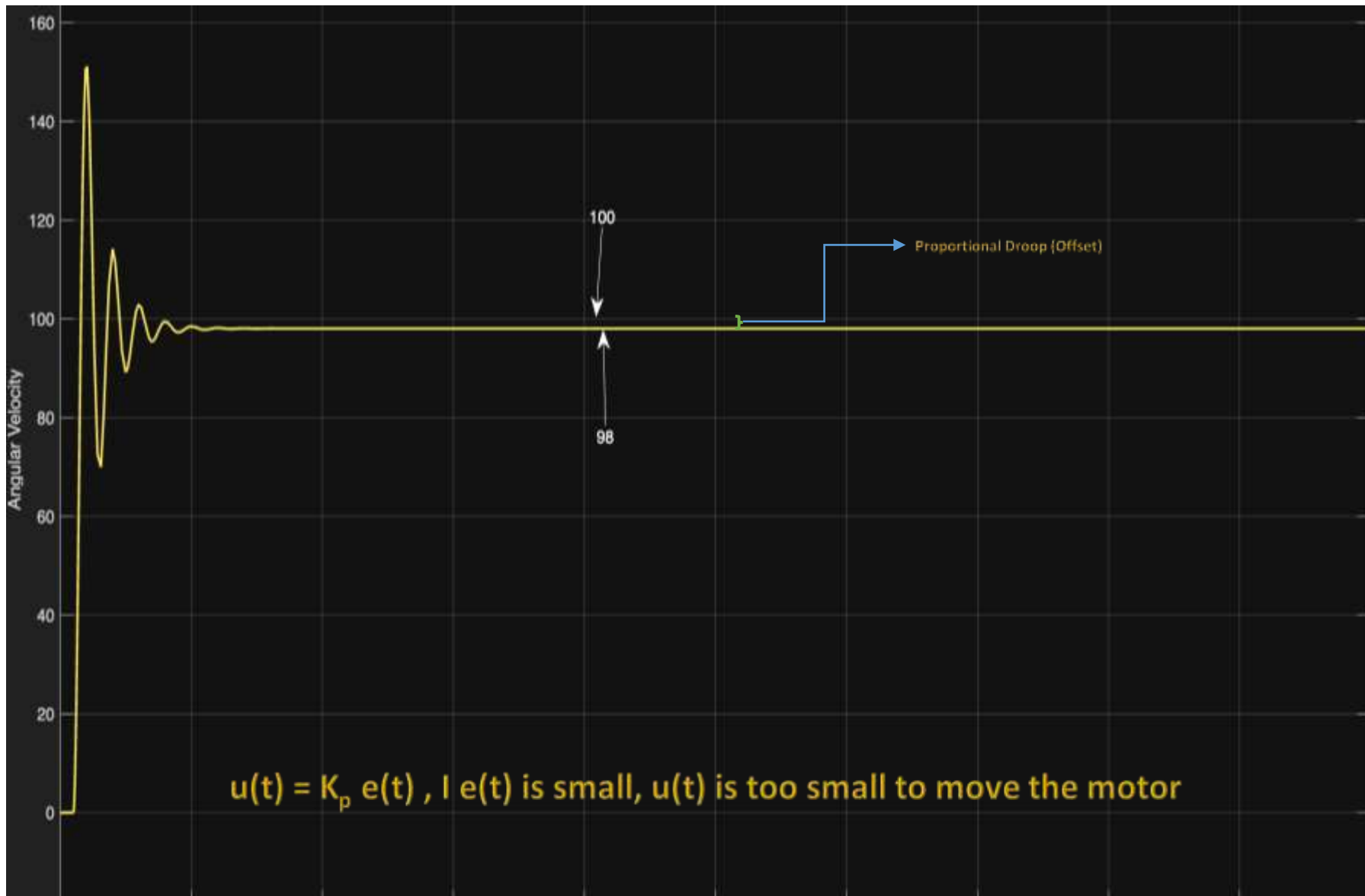
Observations from the P-Loop:

- **Increased Agility:** As K_p increased, the system responded faster compared to the Open-Loop baseline.
- **The Persistent Offset:** Despite high K_p values, the system reached a "plateau" below the 100 rad/s setpoint.

The Scientific Conclusion (The "Why"):

- A Proportional controller requires a non-zero error to generate a control signal.
- If the error becomes zero, the voltage output becomes zero, meaning the motor cannot fight against internal friction (B) and Back-EMF (k_b).

The Necessity of Integral Action: This experiment confirms that the Integral term (K_i) is mandatory to "integrate" the remaining error over time and provide the final voltage boost needed to reach $e_{ss} = 0$.



Heuristic Manual Analysis:

Rising time: $\sim 0.5s$

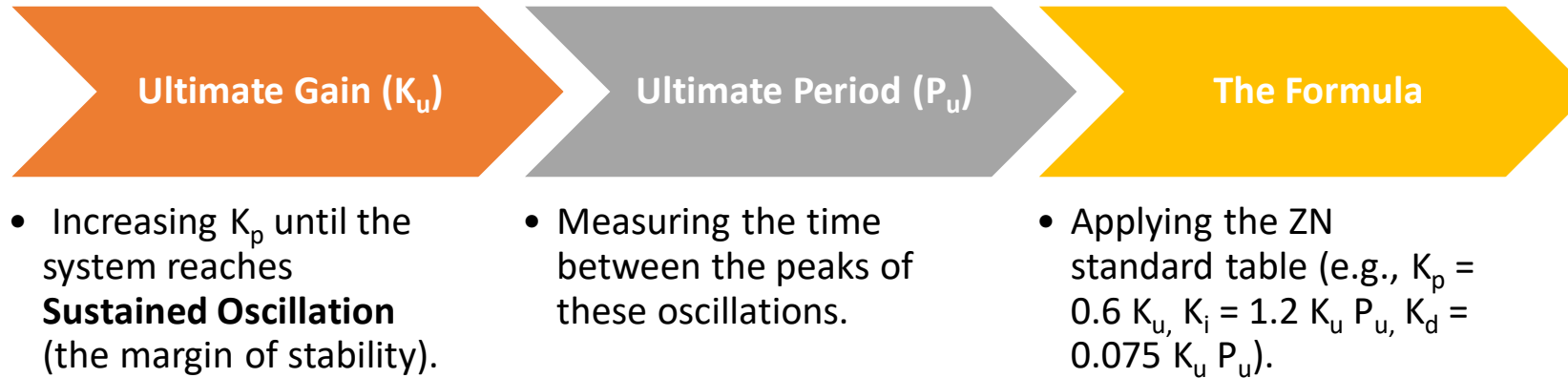
Overshoot: $\sim 5\%$

Settling time: $\sim 2.0s$

Steady-state-error: 2%

▶ Tuning Phase II (Ziegler-Nichols Methodology)

Implementation of the **Heuristic-Analytical** method based on the system's stability limits.

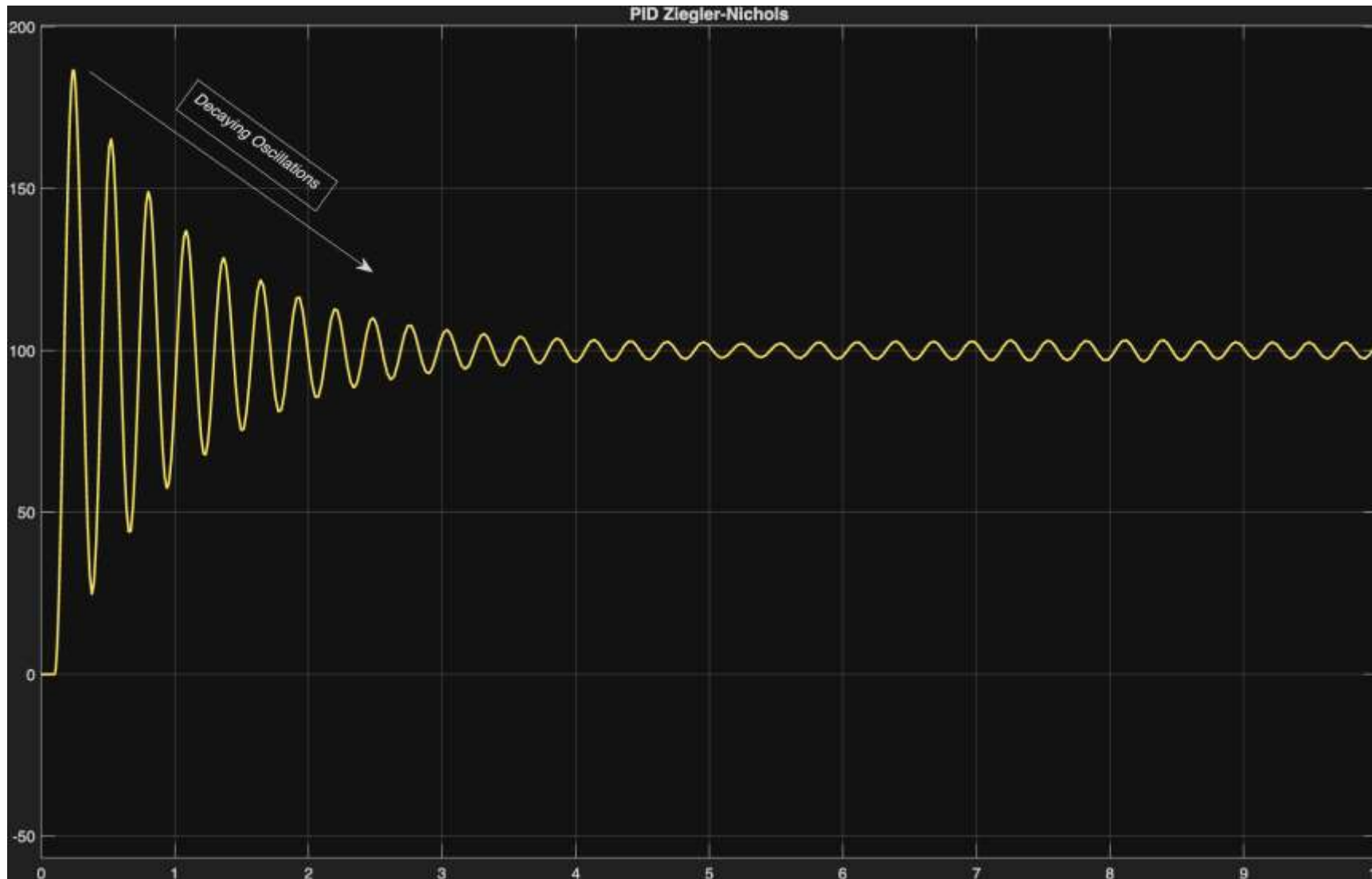


Observations (The Trade-offs):

High Agility: The ZN method typically yields a very fast **Rise Time (T_r)**.

The "Ringing" Effect: Inherently results in high **Maximum Overshoot (M_p)** and decaying oscillations before settling.

While ZN provides a systematic path to zero steady-state error, its aggressive "Quarter-Amplitude Decay" philosophy may be unsuitable for high-precision mechanical loads sensitive to vibration.



Ziegler-Nichols Methodology:

Rise time: ~0.15s
(Fastest)

Overshoot: ~22%
(Highest)

Settling time: ~2.5s

Steady-state-error: 0%

► Tuning Phase III (Auto-Tune Optimization)

Utilization of frequency-response estimation to linearize the non-linear plant at the operating point (100 rad/s).

The Tuning Objective:

Phase Margin Optimization

- Targeting a stability margin (typically 60°) to ensure robust performance against parameter variations.

Balance of Criteria's

- Simultaneously minimizing **Rise Time** while enforcing a constraint on **Maximum Overshoot**.

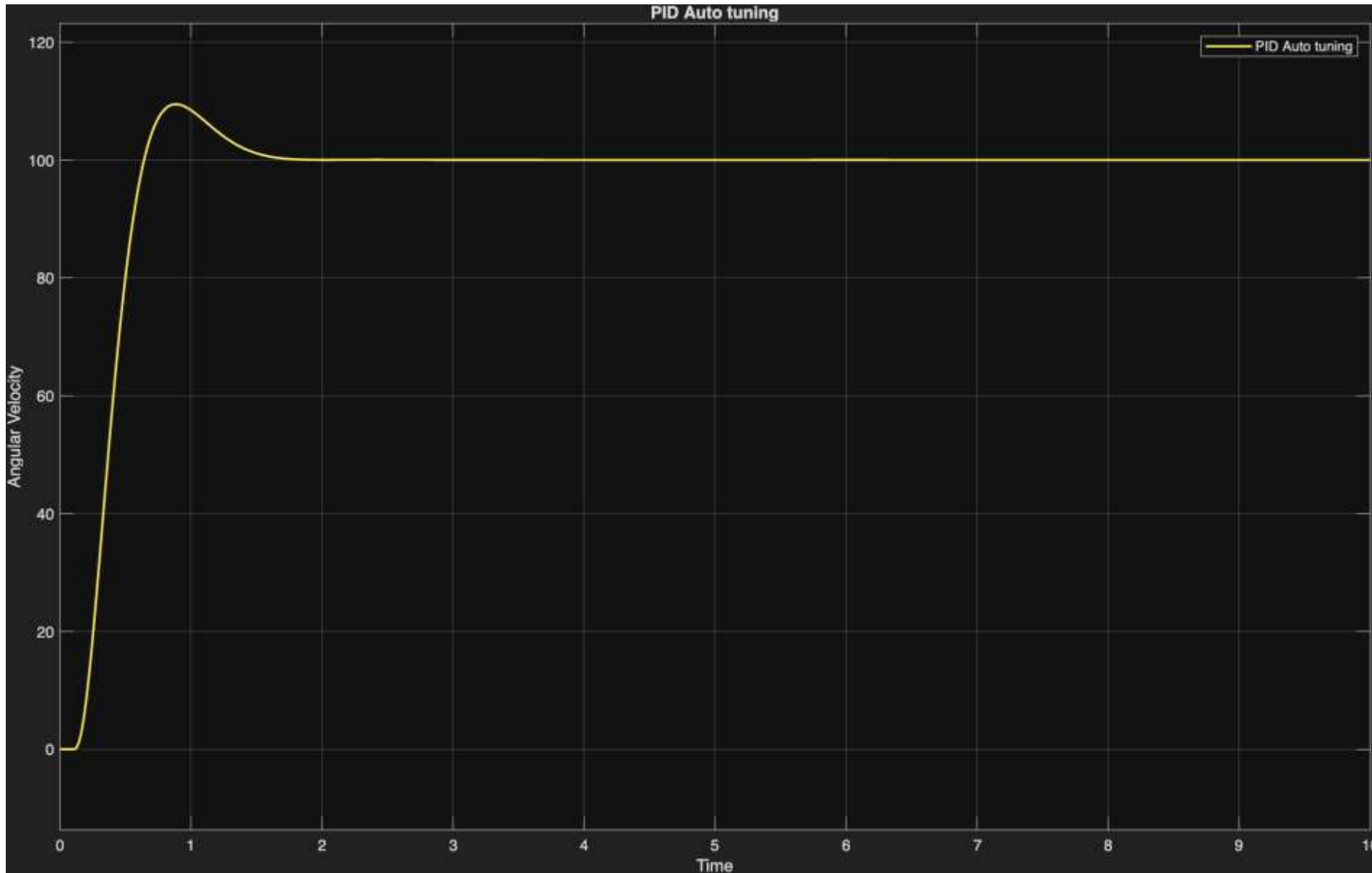
Observed Results:

Critically Damped Behavior

- Achievement of a smooth transition to the setpoint without the "ringing" or excessive oscillations seen in the ZN method.

Stability

- The algorithm accounts for the high-frequency dynamics (Inductance) to prevent "Derivative Kick."



Auto tuning method:

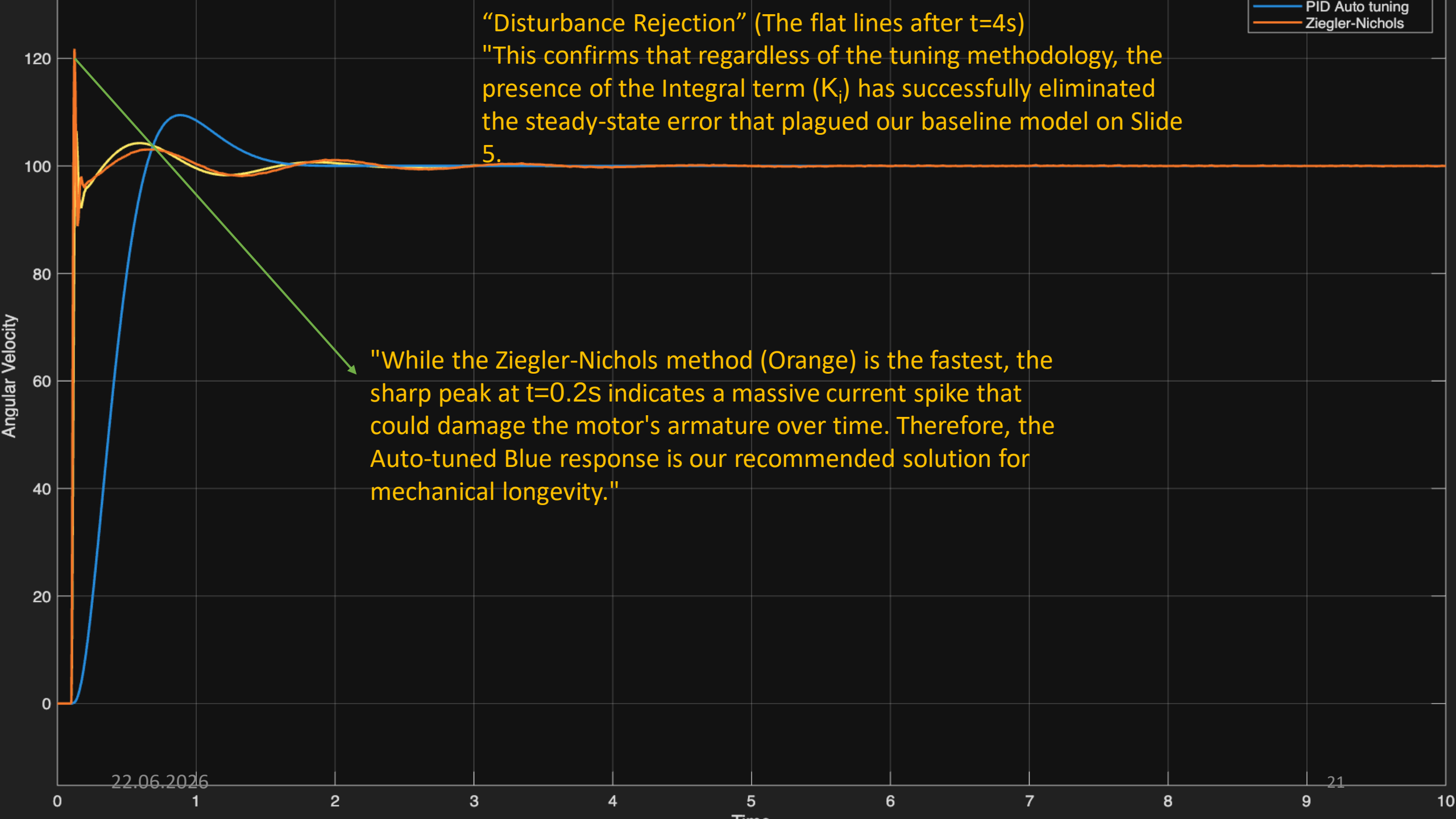
Rise Time: $\sim 0.65s$

Overshoot: $< 10\%$

Settling time: $\sim 1.5s$

Steady-State Error:
0%





REFERENCES:

Step Response with $K_p = 21$, $K_i = 500$ and Different Values of K_d

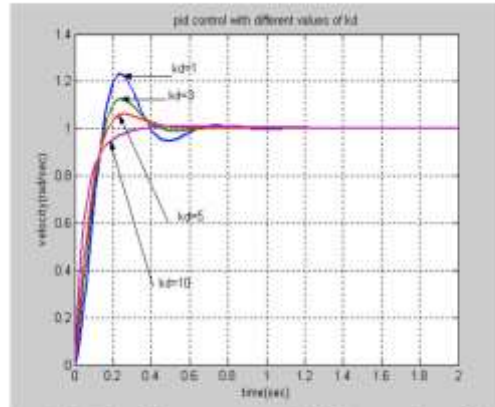
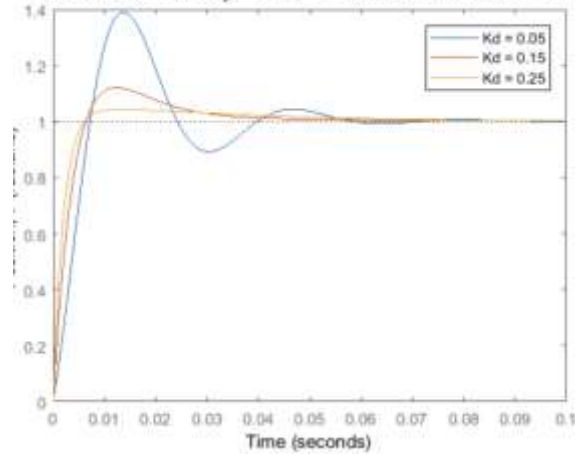


Fig. 6 closed loop response with different values of k_d .

So now we know that if we use a pid controller with: $k_p=100$, $k_i=200$ and $k_d=10$; all our design requirements will be satisfied and the response looks like:

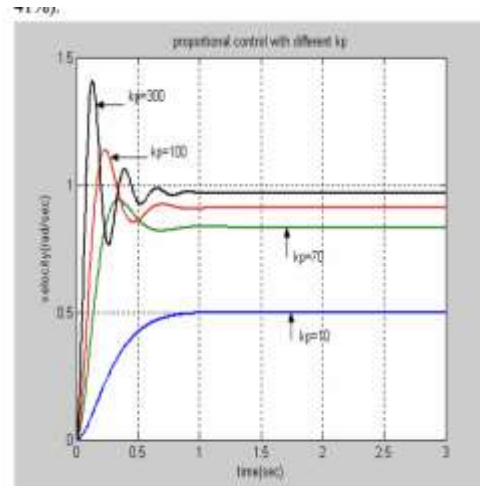


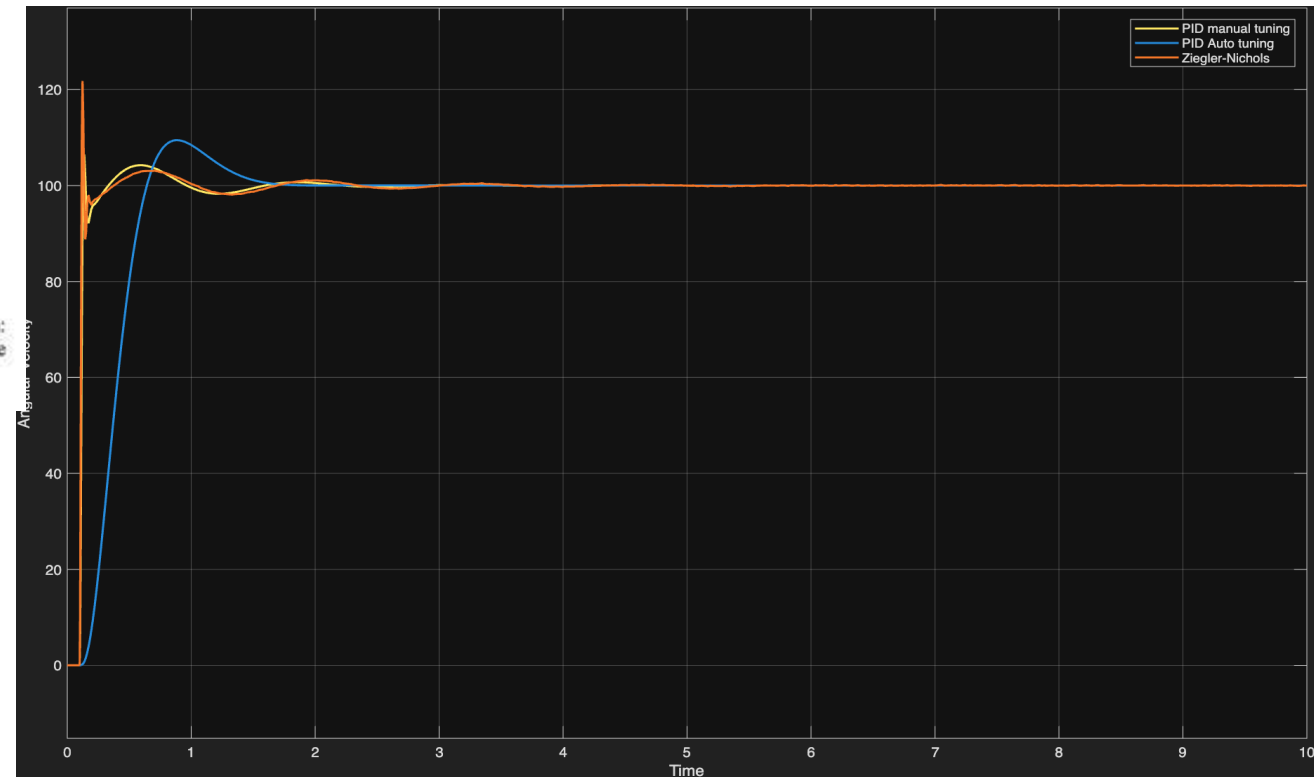
Fig. 4 closed loop response with different values of k_p .

<https://ctms.engin.umich.edu/CTMS/index.php?example=MotorPosition§ion=ControlPID#11>

Proceedings of the 2nd International Conference on Machine Learning and Automation
DOI: 10.54254/2755-2721/81/20241083



Figure 3. Simulation Result 1





Ziegler-Nichols (Orange)

- Notice the "aggressive" spike at the beginning. It reaches the setpoint incredibly fast (lowest Rise Time) but suffers from a significant initial overshoot and high-frequency "chatter." This is the classic ZN trade-off.

Manual Tuning (Yellow)

- You can see this response is conservative. It has very little overshoot but is sluggish compared to the others. It represents the "safe" but non-optimized human approach.

PID Auto-tuning (Blue)

- This is your "Critically Damped" champion. While it is slightly slower to rise than the orange ZN line, look at the smooth transition. It has a controlled overshoot and settles into a perfectly flat steady-state faster than the others.



Comparative Analysis

Metric	Ziegler-Nichols (Orange)	Manual Tuning (Yellow)	Auto-Tune (Blue)
Rise Time (T_r)	~0.15s (Fastest)	~0.5s	~0.65s
Overshoot (M_p)	~22% (Highest)	~5%	~9%
Settling Time (T_s)	~2.5s	~2.0s	~1.5s
Steady-State Error	0%	2%	0%

“This comparison proves that 'optimal' control isn't just about speed; it's about the balance of transient stability and accuracy. With this data, we have successfully validated our Digital Twin's control architecture”.



Conclusion

Synthesis of Research:

Development of a high-fidelity Digital Twin based on first-principles electromechanical laws (KVL/Newton-Euler).

Transition from an unregulated 90% error state to a precision-controlled system via PID.

Final Recommendation:

The **Auto-Tune Optimization** is selected as the primary control architecture.

Justification: It provides the optimal balance of a **60° Phase Margin**, minimal mechanical stress (low overshoot), and the fastest settling time.

Key Engineering Takeaway

The Integral (K_i) term proved essential for **Disturbance Rejection**, ensuring the motor maintains setpoint velocity regardless of external Load Torque (T_L).

Project:

Group members:

Date: 23 -06-2026

Work completed /Status:

- Topic selected: PID Controller Simulation for DC Motor (MATLAB/Simulink)
- Problem statement defined
- DC Motor mathematical model studied (electrical + mechanical equations)
- PID theory reviewed (P, I, D terms and control law)
- Simulink model structure planned
- Tuning methods identified (Manual, Ziegler-Nichols, Auto-tune)
- Concept presentation slides prepared
- Build the DC Motor transfer function in Simulink
- Implement the PID Controller block in closed-loop
- Applied all 3 tuning methods and run simulations
- Recorded step response, settling time, overshoot, steady-state error
- Compared results across tuning methods

Next Steps:

■ Report submission



22.06.2026

Discussion Required:

Project Requirements:

- MATLAB R2023a or later (with Simulink)
- Control System Toolbox
- Standard PC/Laptop (simulation only)

Thank You